Nepal 2026 Flood: Event Reconstruction and Illustrative Loss Workflow

Technical summary for portfolio review

This document stitches the existing modules of the case study. It does not introduce new modelling. The purpose is to present the reconstruction, the infrastructure concentration, the insurance snapshots, the illustrative model output, the risk-financing layers, and the principal data limitations in a single readable form.

1. Event and hazard chain

On 26 August 2026 at approximately 08:37 NPT a rapid high-altitude slope failure occurred in the Langtang Himal near the China–Nepal border. Debris comprising ice, rock and sediment entered the drainage. A debris flood then propagated downstream through the Lhende Khola, Bhote Koshi and Trishuli River system. First reported impacts on the corridor infrastructure occurred within roughly 40 minutes of the initial failure.

The supported sequence is: slope failure → debris enters the drainage → debris flood propagates downstream → rapid impacts on border infrastructure, roads, bridges, settlements and hydropower facilities. A temporary blockage and subsequent outburst remains a hypothesis; it is not treated as confirmed. Source-peak attribution (Langtang Lirung versus Tsangbu Ri) and volume estimates remain unresolved across the available sources and are not adopted as single figures.

The diagram in the timeline module distinguishes evidence status from the physical sequence by marking the blockage step as a hypothesis while keeping the upstream failure and the downstream flood as supported. Detailed source notes and the competing volume estimates are recorded in the event-reconstruction module and are not repeated here.

2. Corridor risk and infrastructure concentration

The affected corridor is a narrow valley that carries the border crossing, roads, bridges, settlements and multiple hydropower projects along the same path. Official figures for Rasuwa district record 41 bridges swept away plus 4 damaged, approximately 41 km of road damaged or destroyed, and 14 hydropower projects with a combined capacity of roughly 748 MW. In Syabrubesi the Copernicus EMSR927 grading product (27 August 2026) reports more than 240 buildings destroyed and 32 damaged.

A full asset-level intersection of exposure with flood depth was not produced. No complete georeferenced inventory exists in the case study, and no event-specific depth layer was generated. The location map shows the pathway and selected hydropower sites; road, bridge and settlement layers are not fully represented. The concentration of assets along a single valley path is therefore documented through official counts and the available grading product rather than through a modelled overlay.

3. Insurance snapshots and the protection-gap question

Two dated Nepal Insurance Authority snapshots are recorded and are not blended. As of 31 August 2026, NPR 25.87 billion had been filed across 583 policies. As of 29 September 2026 the filed total stood at NPR 20.98 billion across 2,180 policies; claim deadlines had been waived and advances of up to half of a preliminary survey were available. Both figures are filed claims, not paid losses. Claims are concentrated in commercial and hydropower assets.

Reconstruction need is substantially larger than the filed-claims total. The difference indicates a protection gap but is not a precise uninsured-loss figure: filed claims are not equivalent to paid losses, and reconstruction need is broader than insured loss. The protection-gap discussion in the insurance module uses the 31 August snapshot only; the 29 September snapshot is recorded separately.

4. Illustrative model output versus national estimate

Running the vulnerability functions on the small synthetic exposure table produces approximately US$33.3 million in ground-up loss under the stated assumptions. The national-loss estimate recorded in the damage totals is approximately US$2.6 billion. The two figures have different scopes: the model result applies only to the included synthetic assets, while the national figure is an aggregate of much broader coverage.

The depth-damage slopes are simplified, literature-informed placeholders drawn from the JRC global flood depth-damage framework (Huizinga, de Moel and Szewczyk, 2017). They are not the published JRC curves reproduced without modification, and they are not calibrated to Nepali construction types or to this event. The curves are not retuned to force agreement with the national estimate. The model output is a deterministic result under the specified inputs; it does not represent a quantified uncertainty distribution.

5. Risk-financing layers

Insurance covers only part of the commercial and hydropower losses. The remaining losses fall on asset owners, lenders and government budgets. Reconstruction need is broader than insured loss. No company-level payout data or donor ledger is available in this case study, so the allocation across layers cannot be quantified from the present sources.

The three layers can be summarised as: (1) the insurance layer (filed claims, partial and concentrated), (2) the private layer (asset owners and lenders), and (3) the public layer (government and reconstruction finance). The illustrative model and the NIA snapshots do not allocate losses across these layers.

6. Data limitations and what would be needed next

The principal limitations are already stated in the data-limitations module and are restated here only in summary form. Exposure data are synthetic and illustrative. Vulnerability functions are not Nepal-calibrated. No hydraulic modelling was performed; flood depths are illustrative inputs. The damage functions do not separately represent the full debris-flood process (sediment, boulder impact, erosion, scour or structural failure). Casualty and damage figures remain provisional and time-dependent. National aggregate totals are used only as contextual checks and are not calibration targets.

A production workflow for a catastrophe-modelling firm and its clients would be strengthened by a georeferenced asset inventory with defensible replacement values, observed or hydraulically modelled flood depths and extents, Nepal-relevant building and infrastructure vulnerability data, event-specific damage surveys, and company- or portfolio-level claims information. Those inputs would allow the same structural sequence—event reconstruction, corridor exposure, insurance comparison, and risk-financing layers—to move from an illustrative demonstration to quantified, client-specific loss estimates and scenario testing.

This summary draws on the existing event-reconstruction, exposure-vulnerability, insurance, and methodology modules. No new calculations or data sources have been introduced.